

Resolution Proof method. (propositional logic)

$\models V$: semantic consequence : always true.

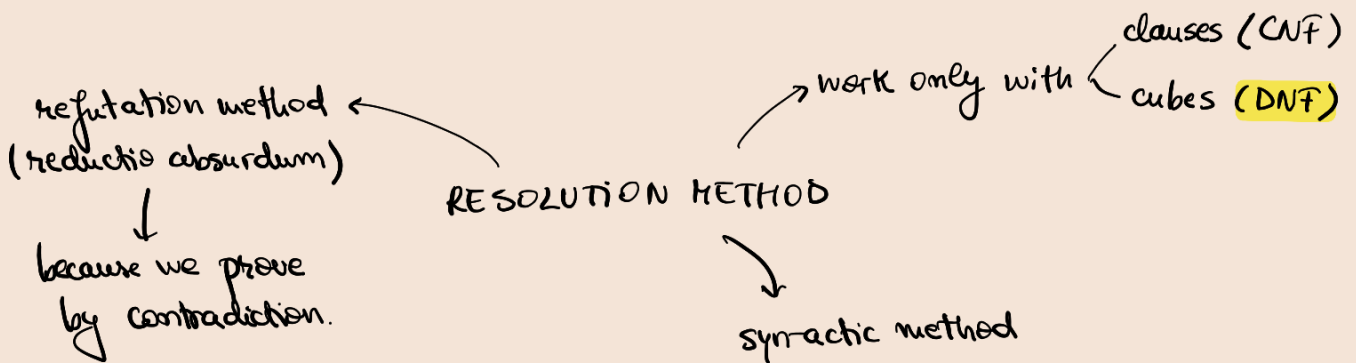
$\models V \Leftrightarrow V$ tautology $\Leftrightarrow V$ is true in every model.

$\vdash V$: syntactic derivability

(V can be derived using $\left\{ \begin{array}{l} \text{inference rules} \\ \text{axioms} \end{array} \right.$)

$\vdash V \Leftrightarrow V$ theorem

In the exam : "Using the refutation / other method,
show that ... is a tautology / theorem".



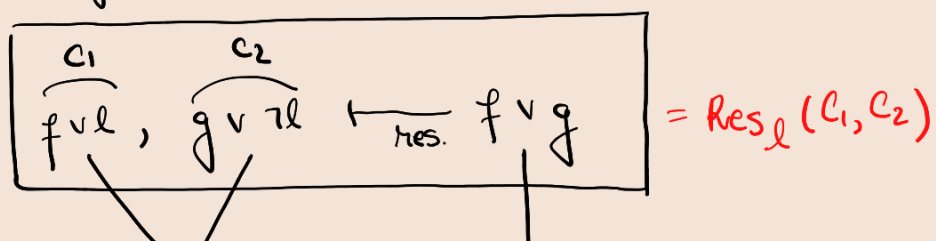
Syntactic approach:

$p \vee q \vee r$ a clause $\left\{ \begin{array}{l} \text{is } \textcolor{green}{T} \text{ when at least one is } T. \\ \text{is } \textcolor{red}{F} \text{ when all are } F. \end{array} \right.$

! always a consistent formula.

How to symbolise inconsistency:

$\square \rightarrow$ the empty clause.



CLASHING
CLAUSES

PARENT
CLAUSE

$$\text{Res}_l(C_1, C_2) = \square \text{ if } \begin{cases} C_1 = l \\ C_2 = \neg l \end{cases}$$

Algorithm: goal $\rightarrow$ check consistency

repetă ... până când

$\rightarrow$ will not be implemented because we need a STRATEGY TO (**) generate all the possible resolvents.

could be generated only
with the algorithm,
but it would be impractical.

we don't need all,
but just enough to derive $\square$

Strategies

set-of-support
level saturation
deletion
linear / lock / semantic

$\rightarrow$ just the necessary resolvents

• Soundness = correctness (never derives false contradictions)

resolution $\rightarrow \square \Rightarrow$ the original set is inconsistent.

• Completeness $\neq$ efficiency

inconsistent set $\Rightarrow$ resolution $\rightarrow \square$

(if the set is inconsistent, the resolution method can derive $\square$, even if inefficient)

• Soundness + completeness

a restricted / poorly chosen strategy may be sound, but not complete because it may miss some necessary resolvents.

$\hookrightarrow$ Similarly, backtracking can get stuck exploring only one branch and never reach the needed contradiction.

Theorem

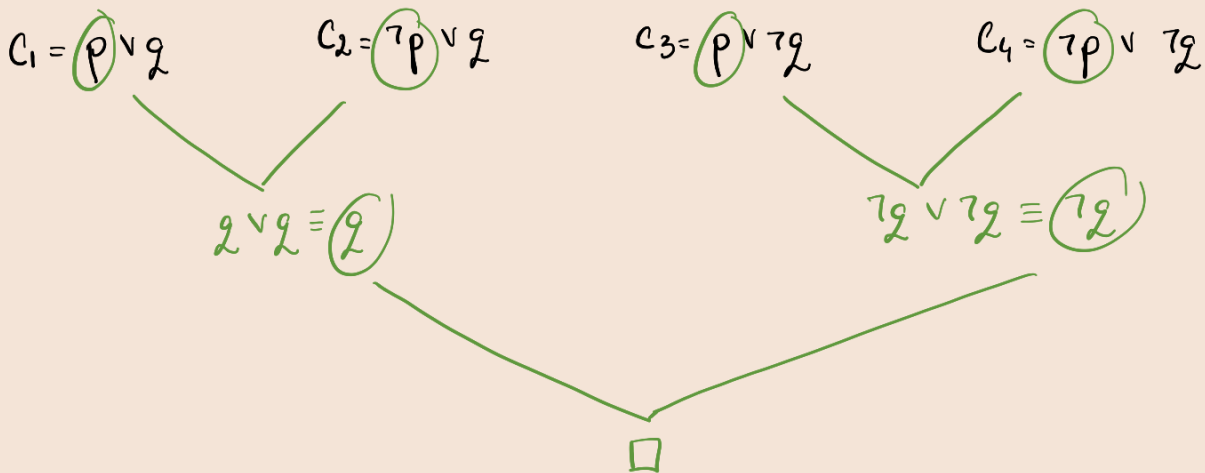
A formula U is a theorem (tautology) $(\Leftrightarrow) \neg U$ is always false.

Pattern:

1. Take $\neg U$.
2. Transform $\neg U$ into CNF.
3. Consider the resulting set of clauses
4. Apply resolution to this clause set.
5. if $\xrightarrow{\text{resol.}} \square \Rightarrow$ the set is inconsistent $\Rightarrow \neg F$ false $\Rightarrow F$ always true $\Rightarrow F$ tautology

F valid $(\Leftrightarrow) \models F (\Leftrightarrow) \neg F$ inconsistent $(\Leftrightarrow)$ resolution gives $\square$.

Ex. 1 Prove that S is inconsistent using Res. Method:
(set)



You could also derive $\square$ using $C_1 - C_3$
 $C_2 - C_4$

! Try to simplify the initial set of clauses by:

removing tautologies ($p, \neg p$)

removing subsumed clauses

$C_1 = p \vee q$ (subsumes C_2)

~~$C_2 = p \vee \neg q$~~

pure literal elimination
(p without $\neg p$)

unit clause propagation

~~$C_1 \Rightarrow p$~~

~~$C_2 = (\neg p) \vee q$~~

Absorption Laws:

$$U \wedge (U \vee V) \equiv U$$

$$U \vee (U \wedge V) \equiv U$$

$$C_3 = A \vee \neg p \vee S$$

$$p \vee q \vee \neg p \equiv T$$

$$\left. \begin{array}{l} \underbrace{p \vee q}_{C_1} \quad \underbrace{p \vee q \vee \neg p}_{C_2} \\ C_1 < C_2 \end{array} \right\} \xrightarrow[\text{laws}]{\text{absorption}} C_1 \text{ subsumes } C_2 \Rightarrow \text{Keep } C_1.$$

Ex.2. $H_1, H_2, H_3 \vdash C \iff \text{CNF}(H_1 \wedge H_2 \wedge H_3 \wedge C) \vdash_{\text{res}} \square.$

! Useful for the optional hw.

Step 1: Translate from natural language into prop. formulas:

$$H_1: S \rightarrow D \wedge A$$

$$H_2: Th \rightarrow B$$

$$H_3: S \wedge Th$$

$$C: Th \wedge B \wedge D$$

Step 2: Transform the implications

$$\boxed{A \rightarrow B \equiv \neg A \vee B}$$

$$H_1: \neg S \vee (D \wedge A) \xrightarrow{\text{distrib.}} \underbrace{(\neg S \vee D)}_{C_1} \wedge \underbrace{(\neg S \vee A)}_{C_2} : \text{CNF}(H_1)$$

$$H_2: \underbrace{\neg Th \vee B}_{C_3}$$

$$H_3: \underbrace{S}_{C_4} \wedge \underbrace{Th}_{C_5} \text{ (already a CNF)}$$

$$C: Th \wedge B \wedge D$$

$$\neg C: \neg (Th \wedge B \wedge D) \equiv \underbrace{\neg Th \vee \neg B \vee \neg D}_{C_6}$$

Step 3: All the clauses

$$C_1: \neg S \vee D$$

$$\text{---} C_2: \text{---} \neg S \wedge A$$

$$C_3: \neg Th \vee B$$

$$C_4: S$$

$$C_5: Th$$

$$C_6: \neg Th \vee \neg B \vee \neg D$$

$$\{C_1, \dots, C_6\} \vdash \frac{?}{res} \square$$

Step 4: Try to simplify the set of clauses:

1) tautologies?

2) subsumed clauses?

3) pure literals?

if yes $\rightarrow$ eliminate those.

Here, C_2 is not useful because $\neg A$ is not found in another clause
 $\downarrow$
 pure literal elimination

Step 5: Apply resolution method. $\Rightarrow$ new clauses.

$$C_7 = Res_{\textcircled{5}}(C_4, C_1) = D$$

$\swarrow$ what we remove

$$C_8 = Res_{Th}(C_5, C_3) = B$$

$$C_9 = Res_D(C_6, C_7) = \neg Th \vee \neg B$$

$$C_{10} = Res_B(C_9, C_8) = \neg Th$$

$$C_{11} = Res_{Th}(C_{10}, C_5) = \square \Rightarrow \text{INCONSISTENCY.}$$

In order to prove inconsistency $\rightarrow$ just find a deriv. of $\square$.

consistency $\rightarrow$ generate all the possible resolvents without deriving $\square$.

$\downarrow$
 we need a STRATEGY.

they are ways to make res. method more efficient.

Level $\rightarrow$ parent:

set 0 (level 0)

$\downarrow$

set 1 (level 1) $\rightarrow$ contains all resolvents of clauses from level 0.

↓
...
Level $\rightarrow$ saturated:

$\hookrightarrow$ when no new resolvents can be generated at that level

Empty set / termination:

$\hookrightarrow$ level saturated, no $\square \Rightarrow$ stop the process.

Ex3. $C_4 - C_1$ (no complementary literals $\Rightarrow$ no resolvent)
 $C_4 - C_2$ (no complementary literals $\Rightarrow$ no resolvent)
 $C_4 - C_3$ (r and $\neg r \Rightarrow$ they resolve upon the literal r)

Lock derivation of the empty clause

$\hookrightarrow$ work with tautologies

cannot be combined with the deletion strategy because the method would not be complete,

deletion strategy removes tautologies, while lock strategy needs them.

Ex: Check the consistency/inconsistency:

$$S = \left\{ \overbrace{\neg r}^{c_1}_{(1)}, \overbrace{p \vee \neg q}^{c_2}_{(2) \quad (3)}, \overbrace{r \vee \neg p}^{c_3}_{(4) \quad (5)}, \overbrace{q}^{c_4}_{(6)} \right\}$$

- r is the literal resolved upon because it has the lowest index.
- we impose an ordering restriction to reduce the nr. of possibilities.

$$C_5 = \text{Res}_{\neg r}^{\text{lock}}(C_1, C_3) = (5) \neg p$$

$\hookrightarrow$ lowest index in C_1 , lowest index in C_3

C_2 does not resolve with C_4 because $\neg q$ does not have the lowest index in C_2 .

$$C_6 = \text{Res}_p^{\text{lock}}(C_2, C_5) = (3) \neg q$$

$$C_7 = \text{Res}_q^{\text{lock}}(C_6, C_4) = \square$$

- We don't know if it's consistent/inconsistent $\Rightarrow$ STRATEGY.

! less resolvents $\Rightarrow$ > efficient.

General resolution with level saturation

The clause set is expanded level by level:

level 0 (orig. clause)



...



level i (all new resolvents generated from level $(i-1)$...)

The level is saturated when all possible resolvents for that level have been produced.

saturation + no new clauses $\Rightarrow$ stop the process.

□ at any level $\Rightarrow$ inconsistency

$$\begin{array}{c} c_1 \qquad c_2 \qquad c_3 \\ \{ \overbrace{p \vee q}^{c_1}, \overbrace{p \vee \neg q \vee \neg r}^{c_2}, \overbrace{\neg p \vee r}^{c_3} \} \Rightarrow S = \emptyset \\ \underbrace{(1) \quad (2) \quad (3) \quad (4) \quad (5)} \quad \underbrace{(7) \quad (6)} \end{array}$$

I can assign indexes
in any order because
I don't have $\neg p$

I will not assign to $\neg p$ the lowest index.

! You might need tautologies to change the relative order.

Linear resolution (graphical level recommended)

The last resolvent will be the parent for the next resolvent.

$\rightarrow$ can be combined with the deletion strategy
because the method would be complete.

Ex. 8 $\rightarrow$ We can choose any clause as a top clause.

$\rightarrow$ As side clauses we can choose any clauses or clauses from the previous iterations.
(input)

